End-to-End Hyperspectral Image Change Detection Based on Band Selection

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***Abstract*— Change detection (CD) aims to identify differences in the same scene at different times. With the increasing amount of hyperspectral images (HSIs), more and more CD techniques use HSIs as the raw data. HSIs often contain redundant bands, where only a few are crucial for CD while others may be detrimental. However, most existing HSI-CD methods extract features directly from full-dimensional HSIs, leading to a degradation of feature discrimination. To tackle this issue, in this article, we propose an end-to-end HSI CD network based on band selection (ECDBS), unlocking the potential synergy between band selection (BS) and CD. The network compromises a deep learning-based BS module and cascaded band-specific spatial attention (BSA) blocks. The BS module selectively retains bands favorable to CD according to the importance of the bands measured based on band correlation. The BSA block tailors the feature extraction strategy for each band based on its feature distribution, allowing extracting sufficient features from each band. Experimental evaluations were conducted on three widely used HSI-CD datasets, demonstrating the effectiveness and superiority of our proposed method over other state-of-the-art techniques.**

***Index Terms*— Attention mechanism, band selection (BS), change detection (CD), deep learning, hyperspectral images (HSIs).**

I. Introduction

HYPERSPECTRAL images (HSIs) are increasingly widely used due to their high spectral and spatial resolution. They typically comprise hundreds of continuous spectral bands in the ultraviolet, visible, near-infrared, and mid-infrared regions of the electromagnetic spectrum. Compared to other remote sensing images, such as visible images, multispectral images, and synthetic aperture radar images, HSIs can capture richer spectral information about ground objects. HSI change detection (CD) aims to recognize differences in multitemporal HSIs taken at the same geographic location. In recent years, HSI-CD has received increasing attention and found broad applications in areas such as forestry and agricultural monitoring [1], natural disaster detection [2], and land surface dynamic analysis [3].

HSI-CD algorithms can be classified into three categories as follows: algebra-based, machine learning-based, and deep learning-based algorithms. The algebra-based methods directly calculate the difference between image pixels, and then determine the changed pixels through selecting an appropriate threshold [4], [5]. However, these methods have high requirements for image preprocessing, and it is difficult to find appropriate thresholds. These deficiencies limit the prevailing of algebra-based CD methods. Machine learning-based approaches use hand-crafted feature extraction methods to convert HSIs into feature space, then contrast the features from bitemporal images [6]. But it is difficult to effectively extract discriminative information through hand-crafted extraction process, resulting in an insufficient performance. In contrast to machine learning, deep learning can extract features more effectively from complex high-level change information. Common neural networks such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and graph convolutional networks (GCNs) have been widely used in HSI-CD. The methods mostly focus on tuning network architecture to improve change feature extraction from full-dimensional HSIs.

However, HSIs inherently contain a lot of redundant and noisy bands, showing a non-negligible impact on CD [7], [8], [9]. It is necessary to mitigate the impact of band redundancy on CD. As an effective dimensionality reduction technique, band selection (BS) can be incorporated into HSI-CD to address this issue. Ou et al. [10] proposed SFBS-FFGNET, which utilizes BS to reduce the dimensionality of multitemporal HSIs to facilitate subsequent CD. However, the approach takes BS as a preprocessing separate from CD, overlooking the synergy between BS and CD.

In this article, we propose an end-to-end HSI CD network based on BS (ECDBS), which exploits the potential synergy between BS and CD. The network mainly consists of a novel deep learning-based BS module and cascaded band-specific spatial attention (BSA) blocks. The BS module aims to select bands that are favorable to CD while removing redundant bands. To do this, it learns to measure the importance of bands based on the correlation between them. Finally, the

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low-dimensional HSI is obtained by clustering all bands and selecting the most important band in each cluster. The BSA block tailors the feature extraction strategy for each band, taking into account the differences in feature distributions of widely spaced bands. It generates a spatial attention map for each band to sufficiently capture change information, improving the discrimination of feature. The main contributions of this article are summarized as follows.

1) We design an ECDBS, aiming to mitigate band redundancy thereby improving HSI-CD performance. The network integrates a novel deep learning-based BS module, which can selectively retain the bands containing sufficient change information, facilitating subsequent change feature extraction.
2) We streamline a pipeline of CD incorporating BS. The proposed HSI-CD network allows end-to-end inference as well as joint optimization of BS and CD, eliminating the need for a separate BS step.
3) To enhance change feature extraction, we propose a BSA block that generates tailored spatial attention map for each band. Different from the widely used uniform feature extraction, the block can extract the features from each band with a strategy tailored to its feature distribution.
4) We perform comprehensive experiments on three widely adopted HSI-CD datasets. The experimental results demonstrate the superiority of our proposed model over other state-of-the-art HSI-CD methods.

II. Related Work

Currently, there have been many HSI-CD algorithms. Generally, they can be categorized into the following three categories, i.e., algebra-based, machine learning-based, and deep learning-based methods. In this section, we briefly describe the three sorts of algorithms in the previous works.

A. Algebra-Based Method

The earliest algebra-based CD methods mainly include image difference method, image ratio method, and image regression method. When applying them to HSI-CD, choosing the optimal band is always a problem. Change vector analysis (CVA) [11] is the most representative algebra-based method. It first generates a difference image by calculating the Euclidean distance between the pixels of two images. Subsequently, the pixels can be classified by comparing the difference and a threshold. But CVA ignores the similarity between adjacent pixels. To overcome this deficiency, Thonfeld et al. [12] proposed robust CVA (RCVA) which considers the influence of neighborhood pixels. Besides, there are several other algebra-based methods [13], such as absolute average difference, vector for angle, and normalized Euclidean distance. These methods typically require a threshold to determine changed and unchanged pixels. The appropriate threshold could be determined manually or by using automatic threshold search methods such as expectation maximization [14] or k-means [15]. Overall, algebra-based methods are easy to implement and valid for easy scene of HSIs. But the accuracy of the radiation and geometric calibration has an important impact on the results of the algebra-based methods.

B. Machine Learning-Based Method

Machine learning-based methods project the HSIs into a new feature space, where the changed features are highlighted and unchanged features are suppressed. The widely used methods include principal component analysis (PCA) [16], multivariate alteration detection (MAD) [17], and slow feature analysis (SFA) [18]. PCA can map data to the direction with the largest variance. PCA with k-means (PCAKM) [6] uses PCA to generate low-dimensional features and perform k-means clustering on the reduced features to obtain CD results. MAD uses canonical correlation analysis [19] to maximize the correlation between the features of multitemporal images. The CD results can be obtained by postprocessing the maximum correlation factor. Nielsen [20] extended MAD to an iteratively reweighted MAD (IR-MAD), which conducts the weighted iteration according to the chi-square distance. SFA extracts the most temporally invariant component from the bitemporal images to transform the data into a new feature space. Multilayer cascade screening strategy (MCS^{4}CD) [21] selects the highly reliable unchanged pixels as training samples and extracts the change information by iterative SFA. Besides, some methods utilize machine learning-based classifier to achieve CD. One type is direct classification method, and the other is post-classification method. The direct classification method is to combine the bitemporal images together, and then a classifier, such as k-nearest neighbors (KNNs) [22] or support vector machine (SVM) [23], is used to find the change categories. The post-classification method first learns and classifies the bitemporary images, respectively, and then compares and analyzes the changes [24]. Although such kind of methods achieve effective dimensionality reduction and noise decline, they need consume an excessive number of computations when dealing with a vast area.

C. Deep Learning-Based Method

Deep learning-based method is currently the mostly used HSI-CD method, as its strong capability of feature learning. A lot of common neural networks have been applied to this field, such as CNNs, RNNs, GCNs, and transformer.

CNNs can fully extract the spatial–spectral features of images; therefore, it is widely used in HSI-CD. Zhan et al. [25] proposed a spectral–spatial convolution neural network with a Siamese architecture (SSCNN-S) for HSI-CD, which employs 1-D convolution to extract the spectral features. Wang et al. [26] proposed SSA-SiamNet that integrates channel and spatial attention in Siamese-CNN to highlight influential information and suppress less informative channels and pixels in the spectral and spatial domains. Gong et al. [27] proposed a spectral and spatial network, which is capable to suppress CD-irrelevant spectral and spatial information via adaptive spectral and spatial mechanisms. Wang et al. [28] designed a residual self-calibrated network that can effectively exploit spatial information. Yang et al. [29] proposed a deep multiscale pyramid network enhanced with spatial–spectral

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residual attention that aggregates low- and high-level features to enrich spatial details and semantic information. Luo et al. [30] fused spatial features from different scales with attention mechanism to generate a more discriminative change feature. Li et al. [31] proposed CBANet that preserves spectral characteristics from high data dimensionality through cross-band feature extraction and 2-D self-attention.

The RNNs use cyclic connections for each neuron activation at each time step, showing great potential in processing continuous multitemporal data. Qu et al. [32] integrated the architecture of encoder–decoder and long short-term memory network (LSTM) to simultaneously learn spatial–spectral features and temporal features. Shi et al. [33] devised a multipath convolutional LSTM that enables the model can learn multilevel temporal dependencies of bitemporal HSIs.

Recently, GCNs, which extend convolution to irregular data structures, have gradually become a research hotspot in the field of HSI-CD. Qu et al. [34] proposed a dual-branch difference amplification GCN, which first introduced GCN to HSI-CD. Wang et al. [35] designed a dual-branch framework (CSDBF) that used CNN to extract pixel-level features and graph attention network to process superpixel-level features. Jian et al. [36] devised a uncertainty-aware graph self-supervised learning framework, which allows encoding structural information of HSIs and data uncertainty into features for downstream CD task.

Moreover, transformer-based HSI-CD models also attracted huge attention due to the advantage of modeling the long-range dependencies. Wang et al. [37] designed a dual-branch transformer for HSI-CD that contains a spectral–spatial transformer and a temporal transformer (SST-Former) and evaluated the model is still valid in small training datasets. Wang et al. [38] devised a triplet transformer framework based on parents-temporal and bother-spatial attention for HSI-CD. Although deep learning-based algorithms have shown significant performance advantages, they still mostly rely on full-dimensional HSIs. This not only limits the potential performance of deep learning models but also reduces the efficiency of training and inference.

D. HSI CD Based on BS

The high-dimensional spectral data of HSIs provides rich information for CD, but also shows significant band redundancy. Influenced by lots of redundant and uncorrelated bands, changes are more implicitly represented and more difficult to detect [7]. High dimensionality also results in high computational burden and huge storage cost [8], particularly in CD task that involves processing multitemporal HSIs. Besides, it may lead to the so-called "curse of dimensionality" or "Hughes phenomenon," i.e., with a fixed number of training samples, the predictive power of a classifier decreases as the dimensionality increases [9]. Consequently, it is necessary to reduce the impact of band redundancy on CD.

BS is an effective dimensionality reduction method to reduce band redundancy. It aims to select a subset of bands from the original HSI and make the low-dimensional data still physically interpretable. Generally, BS methods can be divided into four categories: ranking-based, searching-based, clustering-based, and sparsity-based methods. Ranking-based methods assess the importance of each band according to predefined criteria and choose the top-ranked ones [39]. Searching-based methods treat BS as a combinational optimization problem, which is often solved through heuristic searching strategies [40]. Clustering-based methods first group bands into clusters and then select highly uncorrelated bands from these clusters [41]. Sparsity-based methods regard BS as a optimization problem with sparse constraints and achieve BS through setting the coefficients corresponding to removed bands to zero [42]. Additionally, some new trends in BS are emerging, such as applying graph learning to capture structural information of HSIs [43], leveraging meta-learning to enhance generalization across diverse datasets [44], and associating BS objectives with subsequent analytical tasks [45].

Ou et al. [10] proposed SFBS-FFGNET, which is the first time to introduce the BS technique into HSI CD. It is implemented in two steps: first utilizing a BS method based on SFA to select the significantly changed and unchanged bands, subsequently, using a CNN network to further extract features from the selected bands and produce the final CD result. Although this method achieved remarkable performance, it still views BS as a separate preprocessing from CD, ignoring the synergy between BS and CD. Therefore, we proposed ECDBS, in which the BS is incorporated as an internal unit, to enable the end-to-end inference as well as the joint optimization between BS and CD.

III. Methodology

Given the dual-temporal hyperspectral remote sensing images, we subtract one from the other to obtain the difference image $I \in \mathbb{R}^{B\times h\times w}$, where h and w represent the height and width of an HSI, and B denotes the number of bands. Then, we divide the entire difference HSI I into patches as inputs of ECDBS, denoted $X \in \mathbb{R}^{B\times s\times s}$, where s refers to the length and width of a patch.

As shown in Fig. 1, the ECDBS encompasses a BS module, cascaded BSA blocks, and a multiscale fusion (MF) classifier. The BS module could select a few bands that contribute to CD from all bands, producing a low-dimensional HSI. Subsequently, we extract spatial features from the low-dimensional HSI through cascaded BSA blocks. The blocks can enhance the feature discrimination through adopting the feature extraction strategy tailored to each band. Finally, the features with different scales are integrated and fed into fully connected layers to produce a CD result in the MF classifier. In this section, the three main parts of the proposed ECDBS and the well-designed loss are introduced in the following.

A. BS Module

The high-dimensional and redundant spectral bands of HSI provide a large amount of information for CD, but inevitably increase the difficulty of mining change features. To address this issue, we propose the deep learning-based BS module. It can learn to measure band importance based on the internal correlations between bands. Considering the

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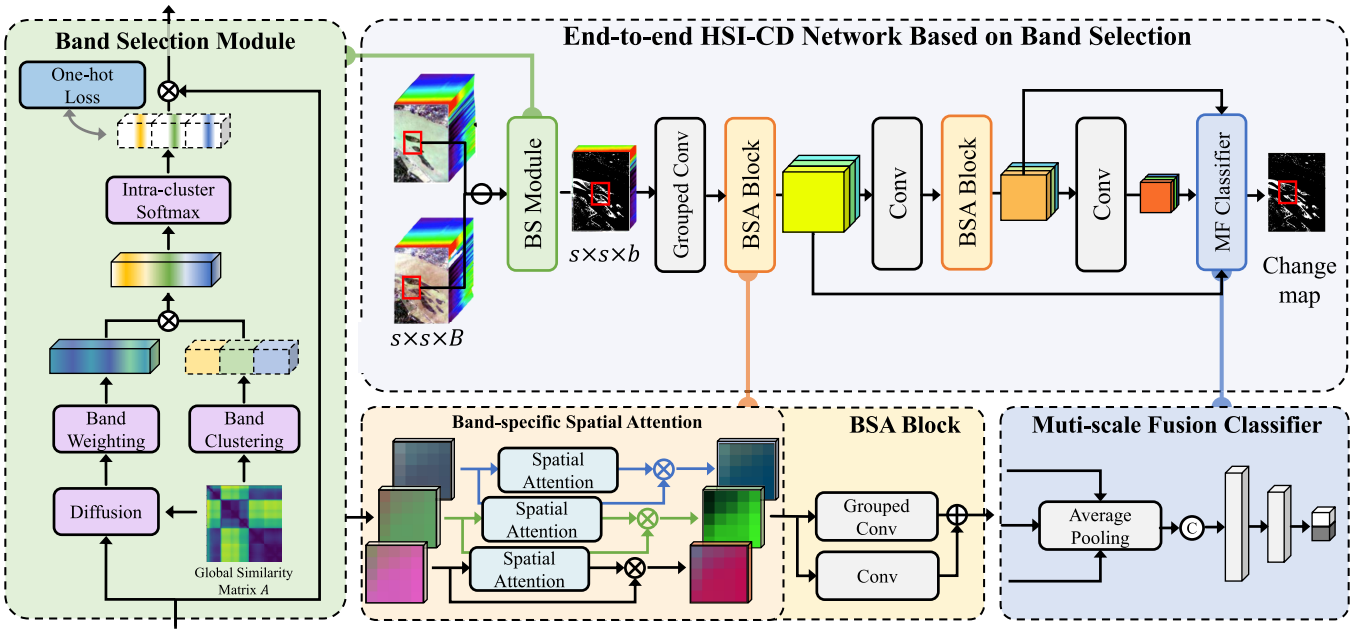

Fig. 1. Architecture of the proposed ECDBS. The ECDBS encompasses a BS module, cascaded BSA blocks, and an MF classifier. We subtract two HSIs to obtain a difference image and subdivide it into patches as input. The patch first undergoes dimensionality reduction in the BS module. Subsequently, the patch with reduced dimensionality is fed into cascaded BSA blocks. Lastly, in the MF classifier, the multiscale features are aggregated and input into fully connected layers to produce the CD result.

amount of information is also important for the selected band subset, we introduce band clustering in the BS to avoid the highly similar bands that are selected, i.e., grouping all bands into a number of clusters and then selecting the most important band from each cluster according to the learned band importance. The selection process contains three steps: first, we adopt a diffusion operation to shrink the difference among patches' band correlation, promoting the learning of consistent band importance weight. Subsequently, the band weighting operation utilizes the band correlation to produce the band importance weights, indicating the amount of each band's contribution to CD. Finally, we apply SoftMax on the band importance weights in each cluster separately. The band with the highest importance weight in each cluster would be selected to form the final band subset. The detailed introduction to the above steps is given as follows.

1) Band Clustering: Band clustering can assign the similar bands into a cluster and relatively different bands into different clusters. Selecting one band from each cluster allows the reserved information is as comprehensive as possible. Therefore, we introduce the band clustering in the BS module to ensure that the selected band subset is still informative while the number of bands is reduced. To this end, we first construct a $B \times B$ similarity matrix A based on a widely used strategy [46], where the similarity between bands is defined as follows:

$$A_{ij} = \begin{cases} \dfrac{e_{i,k+1} - e_{ij}}{\sum_{m=1}^{k} \left(e_{i,k+1} - e_{i,m}\right)}, & I_j \in \mathcal{N}_k(I_i) \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $e_{i,k+1}$ represents the $(k+1)$th smallest Euclidean distance between the ith band I_i and the other bands in I. The Euclidean distance between ith band and jth one is denoted as e_{ij}. $\mathcal{N}_k(I_i)$ refers to the k-nearest neighbors of the band I_i. Using the similarity matrix A, we perform spectral clustering to partition all bands into b clusters. The clustering result is recorded through an assignment matrix $C \in \mathbb{R}^{b \times B}$, where the row of the assignment matrix represents the indexes of the clusters, and the column represents the indexes of the bands. Formally, the assignment matrix is expressed as follows:

$$C_{ij} = \begin{cases} 1, & I_j \in c_i \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where I_j denotes the jth band, and c_i represent the ith cluster.

2) Diffusion: Considering the spatial information of different patches vary widely, it is difficult to learn a consistent band importance weight from it. In contrast, the band correlation of different patches is close to each other. Consequently, the band importance weight will be learned by leveraging the band correlation. To further bring the band correlation of different patches closer, we adopt a diffusion operation. It can be expressed as follows:

$$\overline{X} = \hat{A} X W \tag{3}$$

where W is the learnable parameter, and $\hat{A}$ is normalized from A by $\hat{A} = I - D^{-(1/2)} A D^{-(1/2)}$, where I is an identity matrix and D is the degree matrix of A. In this diffusion operation, each band absorbs the features from other bands according to the similarity defined by the matrix $\hat{A}$. Thus, the band correlation of all patches will approximate the band correlation of the entire HSI, promoting the learning of consistent band importance weight.

3) Band Weighting: Band weighting exploits band correlation information to produce a band importance weight for each band. The importance weight indicates band's contribution to CD. Specifically, we first construct a similarity vector

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$\boldsymbol{s} = \{\boldsymbol{s}_{i\ldots B}\} \in \mathbb{R}^B$ to represent the band correlation as follows:

$$\boldsymbol{s}_i = \sum_{j=1}^{B} \left|\overline{X}_i - \overline{X}_j\right| \tag{4}$$

where Euclidean distance is taken as the similarity metric and $\boldsymbol{s}_i$ is the sum of the Euclidean distances between the ith band and the others. To prevent the biased magnitude of similarity between different patches, we need to normalize $\boldsymbol{s}$ over a batch of patches

$$\hat{\boldsymbol{s}} = \frac{\boldsymbol{s} - \mu_s}{\sigma_s + \epsilon} \odot \gamma + \beta \tag{5}$$

where $\hat{\boldsymbol{s}}$ is the normalized similarity vector, μ_s and σ_s are the mean and the standard deviation calculated per-dimension over a batch of patches, ϵ (e.g., 1e-5) is a constant added for numerical stability, and γ and β are learnable affine transform parameter vectors of size B. $\odot$ is the element-wise multiplication between two vectors. Subsequently, the normalized similarity vector $\hat{\boldsymbol{s}} \in \mathbb{R}^B$ is fed into two fully connected layers to guide the learning of band importance weights $\boldsymbol{w} \in \mathbb{R}^B$. It can be formulated as follows:

$$\boldsymbol{w} = \text{Sigmoid}\big(W_1 \text{ReLU}\big(W_0 \hat{\boldsymbol{s}}\big)\big) \tag{6}$$

where W_0 and W_1 are the learnable parameters.

4) Intra-Cluster SoftMax: To select one band from each cluster, we multiply the band weight $\boldsymbol{w}$ with the assignment matrix C row by row to obtain a weight matrix $\mathcal{W} \in \mathbb{R}^{b\times B}$. Afterward, we perform Softmax on the non-zero elements of each row to highlight one band that contributes most to CD in each cluster, resulting in a selection matrix $E \in \mathbb{R}^{b\times B}$, which is defined as

$$E_{ij} = \frac{\exp\big(\mathcal{W}_{ij}/\tau\big)}{\sum_k \exp(\mathcal{W}_{ik}/\tau)}, \quad \mathcal{W}_{ij} \neq 0,\ \mathcal{W}_{ik} \neq 0 \tag{7}$$

where τ is the temperature parameter that decreases as epoch T increases according to the following rule:

$$\tau = \frac{1}{1 + \log_2 T}. \tag{8}$$

Temperature parameter controls the smoothness of the output distribution. As it becomes smaller, the maximum value of each row vector will become more prominent. Eventually, each row vector of selection matrix will approximate a one-hot vector.

Finally, using the selection matrix, we select the band with the largest importance weight from each cluster. The formulaic representation is as follows:

$$\chi = EX \tag{9}$$

where $\chi \in R^{b\times s\times s}$ is the patch after dimensionality reduction.

B. Band-Specific Spatial Attention

During feature extraction, most CD models uniformly process all bands, ignoring the differences in feature distribution of different bands. This would lead to a degradation of feature discrimination when processing HSIs after BS, as the feature distributions of different bands after BS are significantly different. To address this issue, we design a BSA mechanism that generates tailored spatial attention map for each band based on its feature distribution. Specifically, within each band, we determine the importance of a spatial position by measuring the similarity between its feature and the global spatial feature of the band.

Given the patch feature $\chi = \{\chi_i\}$, $i \in [1 : b]$ where b bands are from different clusters. Each band has a vector representation at every position, namely $\chi_i = \{\boldsymbol{x}_{1,\ldots,m}\}$, where $m = s \times s$, $\boldsymbol{x}_j$ is the spatial vector at position j. For a certain band, we first construct the global spatial vector $\boldsymbol{g}$ through spatial average pooling

$$\boldsymbol{g} = \frac{1}{m}\sum_{j=1}^{m} \boldsymbol{x}_j. \tag{10}$$

With the global spatial vector, the importance coefficient of each position could be calculated. This is achieved by dot product, which measures the similarity between the global spatial vector $\boldsymbol{g}$ and the local spatial vector $\boldsymbol{x}_j$ to some extent. Thereby for each position, we have

$$c_j = \boldsymbol{g} \cdot \boldsymbol{x}_j \tag{11}$$

where $\cdot$ represents the dot product between two vectors. To avoid the magnitude bias among various patches, we normalize the importance coefficient c_j over a batch of patches in the same way as in (5), resulting in a normalized importance coefficient $\hat{c}_j$. Finally, the original $\boldsymbol{x}_j$ is scaled by the generated importance coefficient $\hat{c}_j$ via a Sigmoid function over the space

$$\hat{\boldsymbol{x}}_j = \boldsymbol{x}_j \cdot \text{Sigmoid}\big(\hat{c}_j\big) \tag{12}$$

where $\hat{\boldsymbol{x}}_j$ is the enhanced feature at position j. The enhanced features at all locations collectively constitute the enhanced feature of a band $\hat{\chi}_i$. And the resulted patch feature is composed of the enhanced features from all bands, i.e., $\hat{\chi} = \{\hat{\chi}_i\}$.

As the superior performance of ResNet [47], we embed the BSA and grouped convolution into a residual block, which we refer to as the BSA block, as illustrated in Fig. 1. Following each BSA block, a convolution without padding is applied to increase the receptive field and decrease the output feature map size, which has been widely used in handling patch data with a small size [26].

C. MF Classifier

In CD or visual recognition, integrating features with different scales has been proved to be an effective solution to improve feature discriminativeness and parameter utilizing efficiency [10], [29]. Therefore, we aggregate the features from different depths with different scales to form a comprehensive feature $\boldsymbol{x}_a$

$$\boldsymbol{x}_a = \big[\text{avg}(x_{b1}),\ \text{avg}(x_{b2}),\ \text{avg}(x_{b3})\big] \tag{13}$$

where x_{b1}, x_{b2}, and x_{b3} are the features from the first, second BSA block, and the final convolution layer, respectively, the avg() denotes the spatial average pooling, and $[\cdot]$ represents the concatenation operation. Then the feature is imported to the two fully connected layers to generate the final CD result. Formally, the MF classifier can be expressed as

$$f(\boldsymbol{x}_a) = \text{Softmax}(\Theta_2(\Theta_1 \boldsymbol{x}_a + b_1) + b_2) \tag{14}$$

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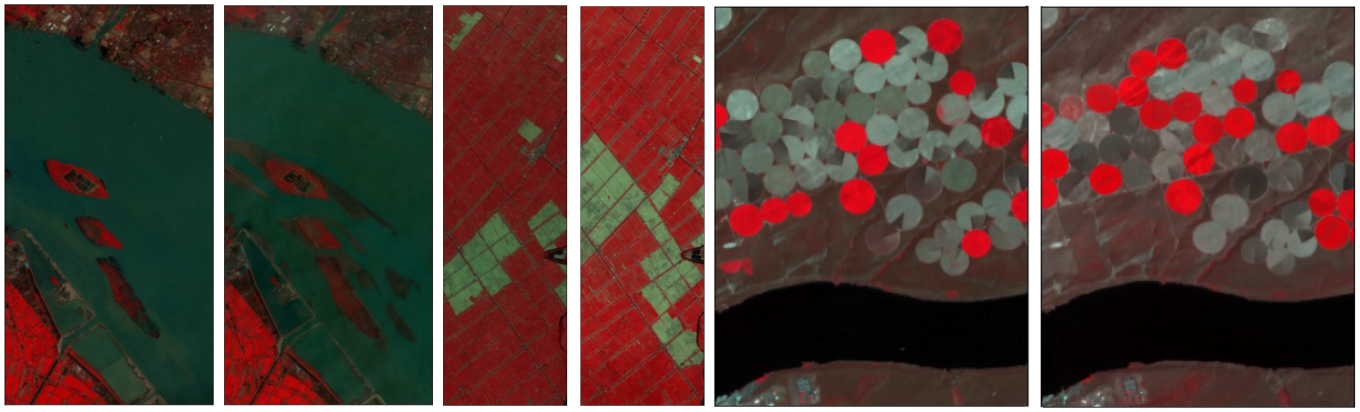

Fig. 2. Three datasets used in experiments. (a) and (b) Are River images acquired on May 3, 2013 and December 31, 2013. (c) and (d) Are Farmland images acquired on May 3, 2006 and April 23, 2007. (e) and (f) Are USA images on May 1, 2004 and May 8, 2007.

where $f()$ denotes the two fully connected layers, and Θ and b are the learnable parameters of the fully connected layers.

D. Loss Function

The loss L for optimizing our model consists of two items, a classification loss L_C for supervising the CD results and a selection loss L_E used to constrain the selection matrix

$$L = L_C + \alpha L_E \tag{15}$$

where α is a trade-off coefficient that balances the minimization between the classification error and constraint item.

We adopt the weighted binary cross-entropy loss as the classification loss to handle the imbalance between the number of changed and unchanged samples, which is characterized as

$$L_C = -\left[\omega_c y \log\left(\hat{y}\right) + \omega_u (1-y) \log\left(1-\hat{y}\right)\right] \tag{16}$$

where $\hat{y}$ represents the predicted probability, y is the true label that is 1 for the changed sample and 0 for the unchanged sample, and ω_c and ω_u are the weights of changed sample and unchanged sample, respectively.

The selection loss aims to guide the row vector of selection matrix E approximates the one-hot vector so that the only one band is selected in each cluster. So, we regularize the entropy of the selection matrix by

$$L_E = -\frac{1}{b} \sum_i^b \sum_j^B E_{ij} \log\left(E_{ij}\right) \tag{17}$$

where E_{ij} is the ith row jth column of selection matrix E.

IV. Experimental

A. Datasets

Our method was evaluated on three widely used HSI datasets: Farmland, River, and USA. These datasets were collected using the Hyperion sensor onboard the EO-1 satellite. The sensor covers wavelengths ranging from 0.4 to 2.5 μm with 242 spectral bands, providing HSIs with a spatial resolution of approximately 30 m and a spectral resolution of approximately 10 nm.

1) River Dataset: The River dataset covers a river region in Jiangsu Province. It includes bitemporal HSIs acquired on May 3, 2013, and December 31, 2013, with a spatial extent of 463 $\times$ 241 pixels. After removing noisy bands, 198 bands are selected for CD. The bitemporal images are shown in Fig. 2(a) and (b) (bands 33, 22, and 11 as RGB). Changes in the dataset are mainly caused by the removal of sediment from the river.

2) Farmland Dataset: The Farmland dataset consists of bitemporal scenes captured over a farmland in Yancheng, Jiangsu Province, China, on May 3, 2006, and April 23, 2007. The dataset has a spatial size of 450 $\times$ 140 pixels, and 155 bands are considered after removing spectral bands with low signal-to-noise ratios. The false-color composites (band 33, 22, and 11 as RGB) of the two images are shown in Fig. 2(c) and (d). Changes in the dataset are mainly caused by crop rotation.

3) USA Dataset: USA dataset was obtained from an irrigated agricultural field in Hermiston City, Umatilla County, OR, USA. The dataset includes two HSIs taken in 2004 and 2007, respectively. This dataset has a spatial size of 307 $\times$ 241 pixels and contains 154 bands after removing noisy bands. The false-color composites (bands 33, 22, and 11 as RGB) of the bitemporal images are shown in Fig. 2(e) and (f). Changes in the dataset are mainly caused by the regulation of irrigation areas.

B. Experimental Setup

1) Sampling Rate and Compared Methods: Referring to the settings in [33] and [35], we randomly select 3.36%, 20.95%, and 9.77% of pixels for training on the River, Farmland, and USA datasets, respectively. Approximately, 1% of these samples are reserved for validation, with the remaining data used for testing. To evaluate the effectiveness of the proposed method, we compare it against other commonly used and advanced approaches, including four classic CNN-based methods: SSCNN [25], PBCNN [48], 3D-CNN [49], and GETNET [50]; an advanced LSTM-based method: CSDBF [35]; as well as two methods that enhances

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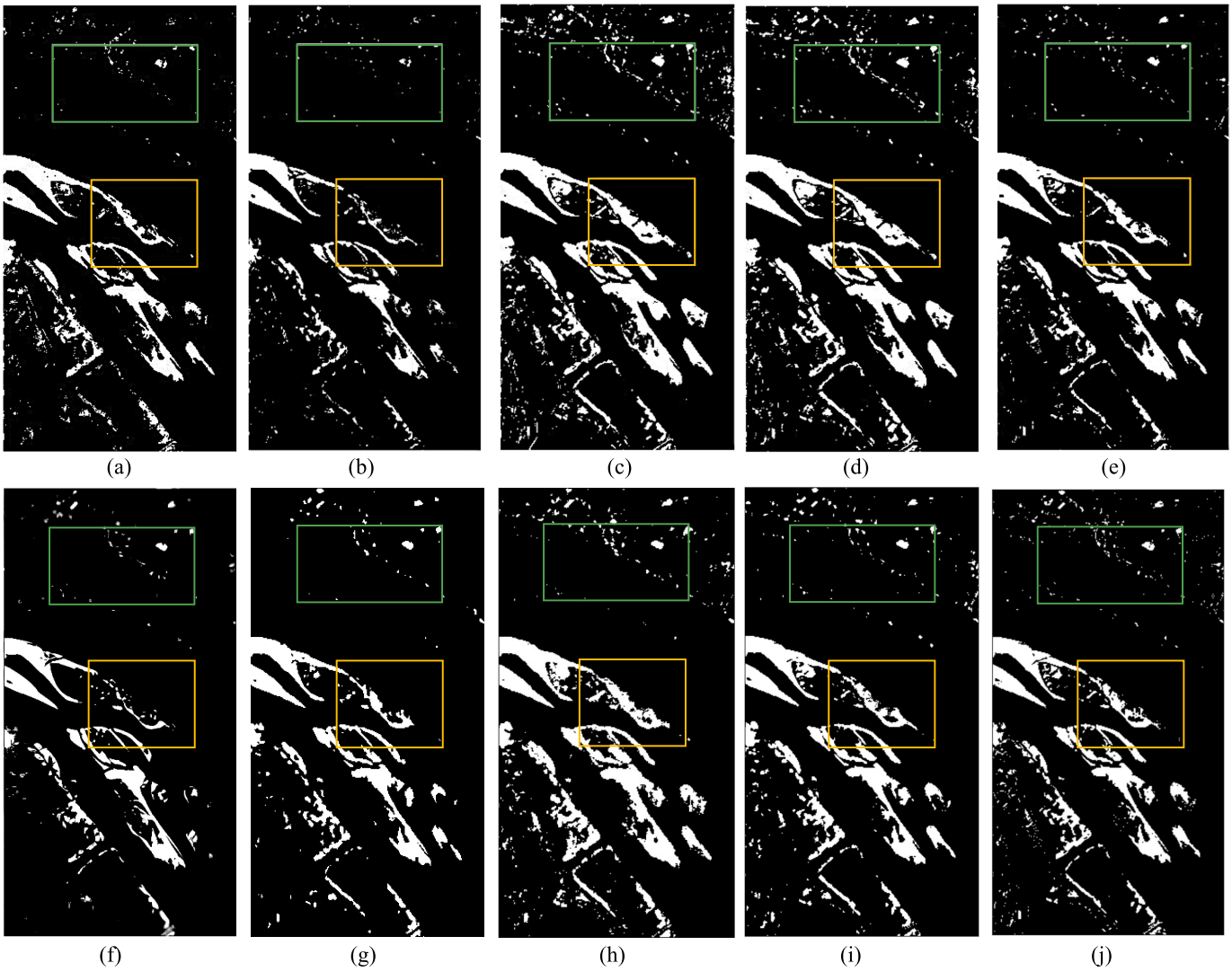

Fig. 3. CD maps for the river dataset. (a) PBCNN. (b) Three-dimensional-CNN. (c) GETNET. (d) SSCNN. (e) SSA-SiamNet. (f) SFBS-FFGNET. (g) CSDBF. (h) CBANet. (i) Ours. (j) Ground truth.

spectral–spatial feature extraction by leveraging attention mechanism: SSA-SiamNet [26] and CBANet [31]. Additionally, SFBS-FFGNET [10], which reduces band redundancy by introducing a BS step separate from CD, is also adopted in the comparison experiment.

2) Evaluation Metrics: To evaluate the performance of different methods, we utilize five commonly used quantitative assessment indices, including overall accuracy (OA), Kappa coefficient (Kappa), precision (Pr), recall (Re), and $F1$-score ($F1$).

OA is the overall accuracy and the proportion of the correctly predicted pixels to the number of pixels in the whole image. The formula is written as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{18}$$

where true positive (TP) denotes the number of correctly detected changed pixels, true negative (TN) represents the number of correctly identified unchanged pixels, false positive (FP) refers to the number of false-alarm pixels, and false negative (FN) indicates the number of missed changed pixels.

Kappa coefficient is an accuracy measure of the classification task. The calculation of the coefficient is based on the confusion matrix, and the value is between -1 and 1. The closer the result is to one, the better the consistency of the classification. Kappa coefficient is calculated as follows:

$$P_C = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN}) + (\mathrm{FN}+\mathrm{TN})(\mathrm{FP}+\mathrm{TN})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2} \tag{19}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - P_C}{1 - P_C}. \tag{20}$$

Precision is the ratio of the number of correctly detected changed pixels to the total number of detected changed pixels. A higher precision indicates fewer FP, meaning the model is better at avoiding incorrectly classifying unchanged pixels as changed. Precision is calculated as

$$\mathrm{Pr} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{21}$$

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TABLE I
CD RESULTS IN THE RIVER DATASET, MODEL PARAMETERS, AND FLOPS

Method	OA(%)	Kappa(%)	F1(%)	Pr(%)	Re(%)	Parameters(k)	FLOPs(k)
PBCNN(2017)	96.02	71.61	73.79	75.11	72.62	235.17	3416.93
3D-CNN(2017)	96.71	73.76	75.39	95.31	62.21	245.79	492.23
GETNET(2019)	95.51	74.42	76.76	65.34	92.84	154179.42	-
SSCNN-S(2021)	95.54	74.08	76.51	64.95	93.25	21.68	1082.72
SSA-SiamNet(2022)	97.35	82.01	83.36	82.26	84.50	107.18	1795.66
SFBS-FFGNET(2022)	96.70	77.10	82.09	84.33	79.86	338.18	5445.22
CSDBF(2022)	96.97	79.82	82.31	84.66	80.10	122.41	-
CBANet(2023)	96.73	75.60	78.50	70.12	89.18	343.35	6791.19
Ours	97.46	83.15	84.53	80.87	88.54	42.21	288.64

Recall is the ratio of the number of correctly detected changed pixels to the number of all actually changed pixels. A higher recall indicates that the model is better at capturing all the changed pixels, even if it might also include more FP. Recall is calculated as

$$\mathrm{Re} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{22}$$

$F1$-score is the harmonic mean of the precision and recall, so this criterion can balance the impact of precision and recall and reflect the average performance of the model. When $F1$ is higher, it indicates that the CD method is more effective. The formula can be written as follows:

$$\mathrm{F1} = 2 \times \frac{\mathrm{Pr} \times \mathrm{Re}}{\mathrm{Pr} + \mathrm{Re}}. \tag{23}$$

3) Parameter Setting: In the proposed method, several hyperparameters, such as the number of selected bands, the number of channels in feature map, and the patch size, can affect the model's performance in the final CD task. Hence, we analyzed the impact of these parameters in our hyperparameter analysis experiments, which is shown in Section IV-F. In comparison experiments, we select 12, 10, and 10 bands from the 198, 155, and 154 spectral bands in the River, Farmland, and USA datasets, respectively. The number of channels in feature map is set to three times the number of selected bands, i.e., 36, 30, and 30 for the River, Farmland, and USA datasets, respectively. For the trade-off parameter α, we set it as 0.1 empirically. Considering the number of changed samples is much less than that of unchanged sample, we set the weights ω_c and ω_u of changed samples and unchanged sample to 5 and 1, respectively, to increase the penalty for changed samples. Furthermore, the hyperparameter settings of the comparable methods are kept consistent with those described in their respective articles. The input patch size is 5×5 for all methods except for GETNET and CSDBF which take the entire HSI as input. To avoid biased estimation, all experiments are carried out using Pytorch in a single NVIDIA RTX 3060 GPU with 12-GB memory.

C. Experiment on River Dataset

Table I displays the results of testing all methods on the River dataset, where the proposed method achieves the best performance in terms of OA, Kappa, and $F1$ indices. Due to class imbalance in the dataset, the OA values of most methods are close, but significant differences can be observed in Kappa values. The proposed method demonstrates a clear advantage in the Kappa index and achieved a 1.14% improvement over the second-best method, SSA-SiamNet. Different

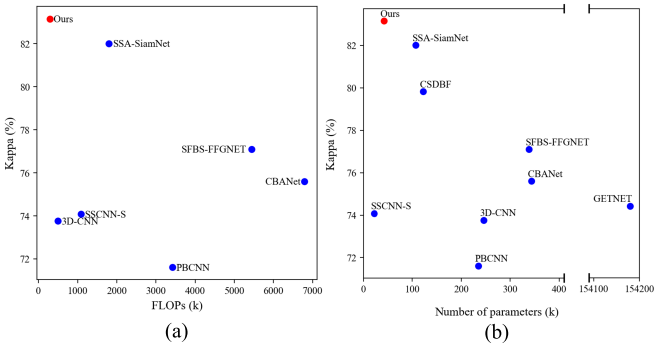

Fig. 4. Comparing performance versus model complexity in River dataset. (a) Performance versus FLOPs. (b) Performance versus number of parameters.

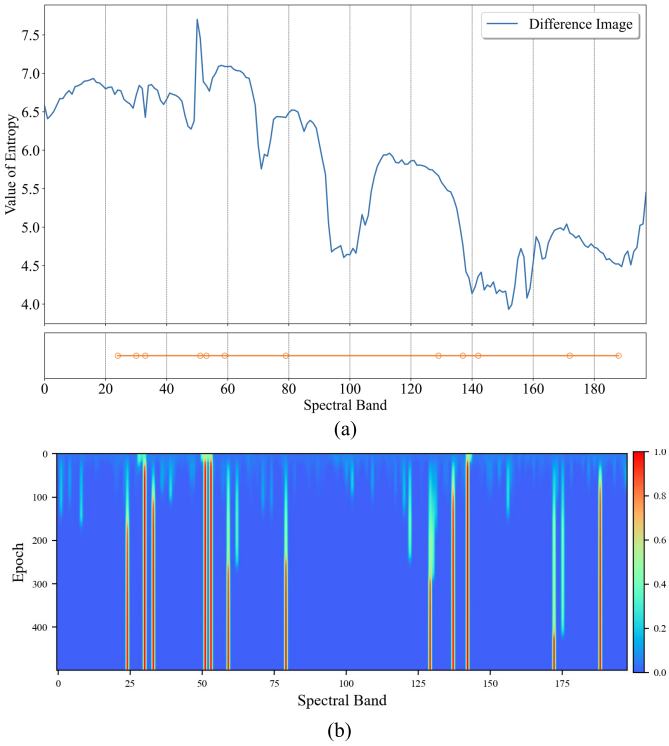

Fig. 5. Analysis of BS result in River dataset. (a) 12 bands selected by our model (below) and the entropy value of each band (above). (b) Visualization of band importance weights across different epochs.

from SFBS-FFGNET, which employs a BS method that is totally independent with the subsequent CD model, our model integrates a BS module that is trained jointly with the CD model. It allows our model to select more discriminative bands for CD, showing better performance.

Fig. 3 presents the CD maps obtained by the different methods. For most methods, it can be observed that many small objects are misidentified over the green box and the CD results in the yellow box do not match well with the ground truth. However, the CD map obtained by our model is the closest to the ground-truth map.

For evaluating the trade-off between the complexity and accuracy of each model, the number of trainable parameters and the floating point of operations (FLOPs) of different models are also provided and listed in the last two columns of Table I. For fairness, we do not compare FLOPs with

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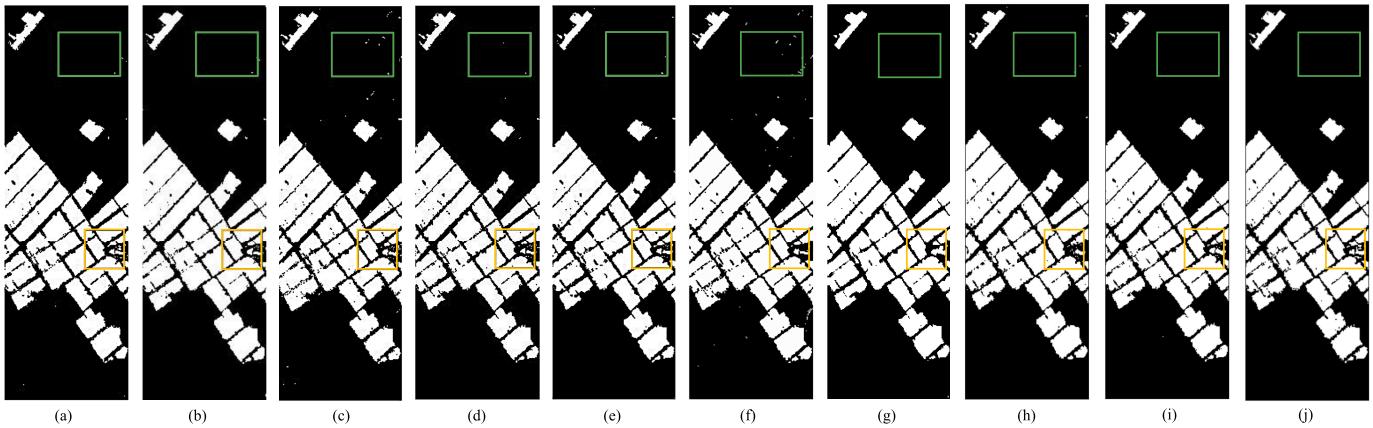

Fig. 6. CD maps for the Farmland dataset. (a) PBCNN. (b) Three-dimensional-CNN. (c) GETNET. (d) SSCNN. (e) SSA-SiamNet. (f) SFBS-FFGNET. (g) CSDBF. (h) CBANet. (i) Ours. (j) Ground truth.

GETNET and CSDBF, because both methods inference with an entire image and typically cause higher FLOPs. It can be seen intuitively in Fig. 4 that our method has fewer parameters and computational requirements, while achieving superior performance compared to other methods.

The 12 bands selected by BS module and the entropy of difference image are shown in Fig. 5(a). We can see the 12 bands are carefully selected, avoiding adjacent bands containing high redundancy. It is attributed to clustering constraint that effectively partitions all bands into several groups with low correlation. The convergence process of band importance weight is illustrated in Fig. 5(b). During the training process, those bands that are favorable to CD are progressively highlighted, while bands that do not aid in CD are suppressed. After 400 epochs, the 12 bands that are most beneficial to CD in all bands are identified and the band importance weights become stable.

D. Experiment on Farmland Dataset

Table II presents the results of all experimental methods on the Farmland dataset. Our proposed method achieves the highest OA, Kappa, and $F1$, with increases of 0.13%, 0.34%, and 0.27% respectively, over the second-best method, SSCNN-S. CSDBF also performs well, achieving the third-best performance with an OA of 98.43%, a Kappa coefficient of 96.20%, and an $F1$-score of 97.06%. Compared with SSA-SiamNet that employs one attention map to scale spatial features of all bands, our proposed model shows a significant improvement with a 0.46% increase in OA and a 0.66% increase in Kappa index. The possible reason is that our proposed model considers the differences in feature distributions among bands while feature extraction rather than uniformly treat all bands.

The change maps derived by different CD models are shown in Fig. 6. It can be observed that most methods display varying degrees of false detection for small objects in green box, but our model, CSDBF, and CBANet show more accurate classification in this regard. Furthermore, the CD results of our method in the yellow box match better with the ground truth.

TABLE II
CD RESULTS OF THE FARMLAND DATASET, MODEL PARAMETERS, AND FLOPS

Method	OA(%)	Kappa(%)	F1(%)	Pr(%)	Re(%)	Parameters(k)	FLOPs(k)
PBCNN(2017)	97.61	94.13	95.79	95.01	96.59	210.40	2797.73
3D-CNN(2017)	98.01	95.20	96.54	94.79	98.33	196.25	393.16
GETNET(2019)	97.79	94.46	95.99	94.39	97.74	93415.26	-
SSCNN-S(2021)	98.45	96.13	97.19	96.69	97.44	18.07	902.12
SSA-SiamNet(2022)	98.12	95.81	96.34	96.24	96.44	88.60	1486.06
SFBS-FFGNET(2022)	97.60	94.30	95.94	93.36	98.66	338.18	3295.40
CSDBF(2022)	98.43	96.20	97.06	97.18	96.94	105.64	-
CBANet(2023)	98.03	95.11	96.47	96.51	96.42	332.34	6511.19
Ours	98.58	96.47	97.46	97.43	97.48	30.68	213.64

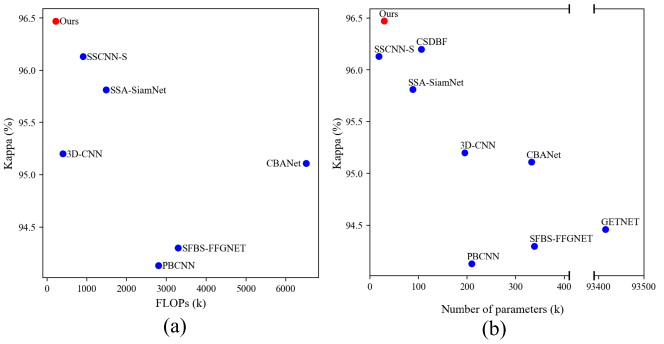

Fig. 7. Comparing performance versus model complexity in Farmland dataset. (a) Performance versus FLOPs. (b) Performance versus number of parameters.

Analyzing the data in the last two columns of Table II, our proposed method uses fewer parameters to achieve better performance in CD and possesses the least computational overhead. As can be seen intuitively in Fig. 7, our model strikes a good balance between complexity and CD performance. This superiority can be attributed to our proposed BS module and the BSA. The former eliminates spectral redundancy of HSIs so that our model does not need extra parameters to process the irrelevant bands. The latter improves the discrimination of changed features in the selected bands, enabling the accurate CD based on low-dimensional HSIs.

The distribution of selected ten bands and the entropy of differential HSI are presented in Fig. 8(a). As we can see, the selected bands not only distribute uniform but also avoid the

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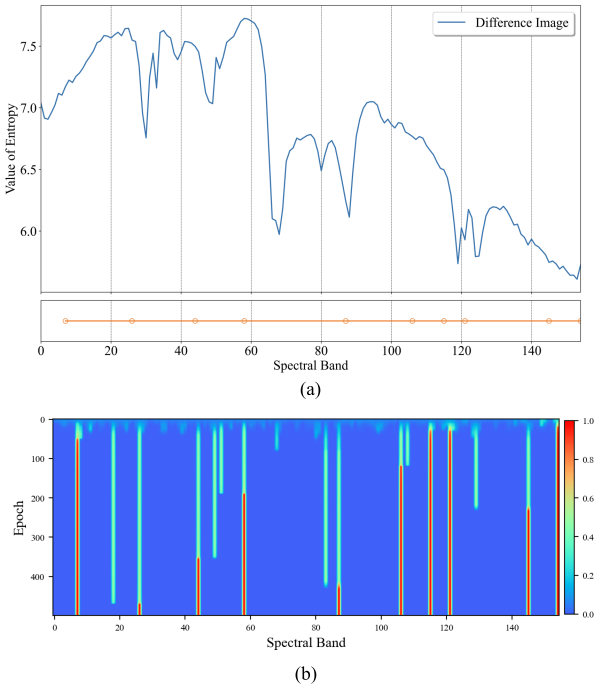

Fig. 8. Analysis of BS result in Farmland dataset. (a) Ten bands selected by our model (below) and the entropy value of each band (above). (b) Visualization of band importance weights across different epochs.

TABLE III
CD RESULTS OF USA DATASET, MODEL PARAMETERS, AND FLOPS

Method	OA(%)	Kappa(%)	F1(%)	Pr(%)	Re(%)	Parameters(k)	FLOPs(k)
PBCNN(2017)	95.72	86.80	89.43	88.32	90.55	209.83	2783.33
3D-CNN(2017)	96.31	88.25	90.59	91.54	89.77	195.10	390.86
GETNET(2019)	93.97	82.41	86.26	79.41	94.45	93415.26	-
SSCNN-S(2021)	96.30	88.64	91.03	87.48	95.06	17.98	897.92
SSA-SiamNet(2022)	95.80	87.41	90.01	86.71	93.37	88.22	1478.86
SFBS-FFGNET(2022)	95.99	88.25	90.82	92.17	89.50	338.18	3295.40
CSDBF(2022)	96.87	90.93	92.70	94.62	90.80	105.25	-
CBANet(2023)	96.64	90.01	92.15	91.84	92.47	332.08	6506.39
Ours	96.95	90.81	92.74	94.28	91.24	29.19	199.30

most sharply decreasing regions with low entropy. And the convergence process of band importance weights is shown in Fig. 8(b).

E. Experiment on USA Dataset

Table III presents the CD results of the different methods in the experiment on USA dataset. Our method achieves the highest OA and $F1$, demonstrating better overall performance than other methods. Although CSDBF gives the highest Kappa, it is slightly inferior to our method in other metrics and contains five times the number of parameters as ours. CBANet also demonstrates good performance, achieving the third-best results with an OA of 96.64%, a Kappa coefficient of 90.01%, and an $F1$-score of 92.15%. Moreover, we observe that SSCNN-S does not achieve great performance with a lightweight network structure. The reason may be that it directly applies the 1×1 convolution layer to fuse the redundant bands to reduce the spectral dimension of HSI, making it difficult to obtain discriminative low-dimensional features in a limited sample number. We suppose that a deep learning-based BS module is a more sensible option to reduce the spectral dimension of HSI, because it can avoid the interfere from irrelevant bands and select discriminative bands. Fig. 9 presents the change maps obtained from different methods. As we can see, most methods fail to accurately detect the changed pixels in the center of green box. Moreover, a segment of the riverbank in the yellow box that has not changed is incorrectly detected as a change by most methods. Our method shows better performance in the two areas. From Fig. 10, we could further observe that our model not only has excellent CD performance, but also shows superiority in terms of computational overhead and number of parameters. The distribution of selected ten bands and the entropy of differential HSI are shown in Fig. 11(a). And the convergence process of band importance weights is presented in Fig. 11(b).

TABLE IV
CD RESULTS USING DIFFERENT NUMBERS OF SELECTED BANDS AND NUMBERS OF CHANNELS IN FEATURE MAP

Dataset	Number of selected bands	Number of channels in feature map	OA	Kappa	F1
River	6	12	97.27	82.09	83.57
	6	18	96.89	80.73	82.41
	12	24	97.06	81.22	82.82
	12	36	97.46	83.15	84.53
Farmland	5	10	98.33	95.83	96.99
	5	15	98.53	96.35	97.37
	10	20	98.44	96.12	97.20
	10	30	98.58	96.47	97.46
USA	5	10	96.38	88.92	91.2
	5	15	96.40	89.00	91.26
	10	20	96.44	89.17	91.41
	10	30	96.95	90.81	92.74

F. Hyperparameter Analysis

The performance of ECDBS relies on the quality of the hyperparameters. Therefore, the influence of number of selected bands, number of channels in feature map, and patch size are investigated.

1) Number of Selected Bands: We tested two different band downsampling factors 32 and 16 to analyze the impact of number of selected bands on our model. As shown in Table IV, the model performance suffers a more or less decline when the selected bands are less. Especially in River and USA datasets, there is an obvious declining trend, which reveals that 1/32 total bands are not enough to provide sufficient change information to achieve excellent CD. We did not further reduce the band downsampling factor to 8 because it would result in an excessive number of retained bands, leading to an increased model complexity. To effectively reduce band redundancy and maintain a low model complexity, we finally decided to set the downsampling factor to 16, i.e., 12, 10, and 10 bands were selected from the River, Farmland, and USA datasets, respectively.

2) Number of Channels in Feature Map: We experimented with channel expansion rates 2 and 3 to find an appropriate number of channels in feature map. In Table IV, we could observe that a smaller number of channels typically leads a worse performance. It may be because the reduction in parameters caused a decrease in the network's ability to express complex features. We did not further increase the channel expansion rate to 4 because it would raise the number

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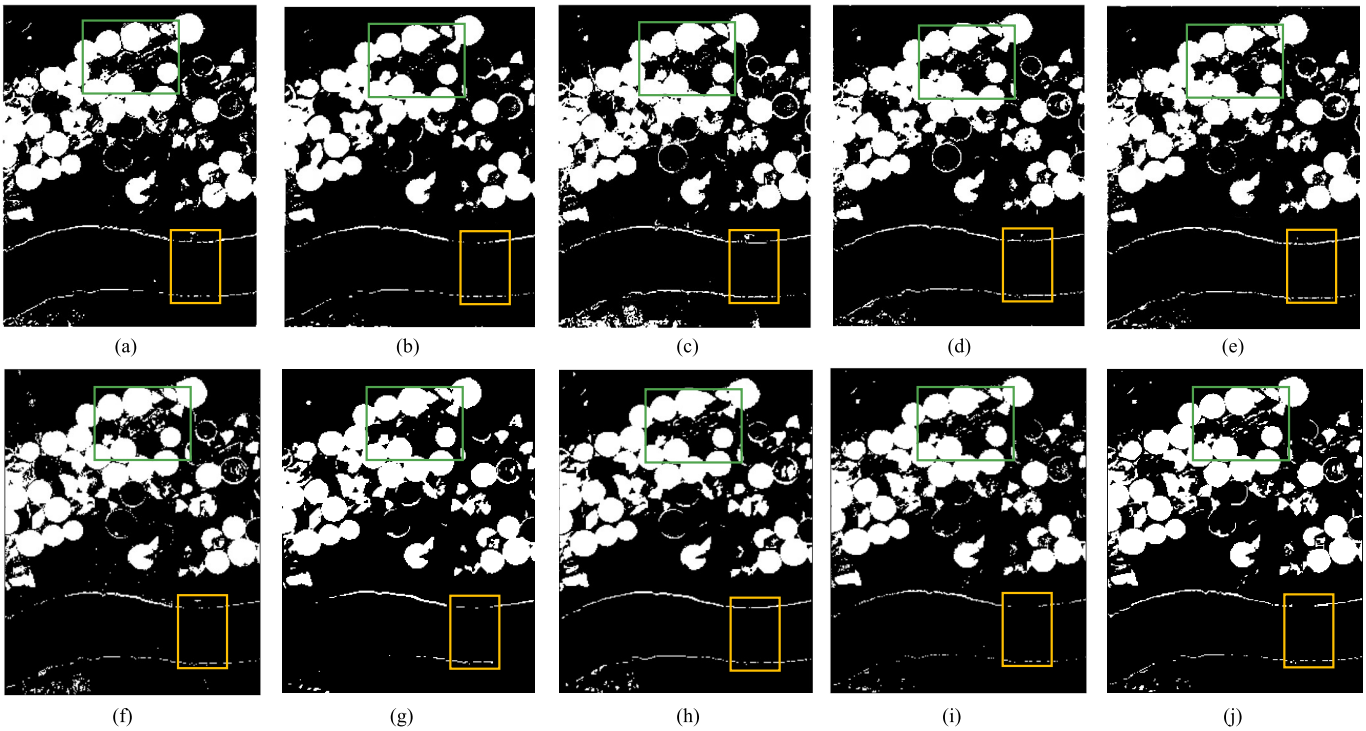

Fig. 9. CD maps for USA dataset. (a) PBCNN. (b) Three-dimensional-CNN. (c) GETNET. (d) SSCNN. (e) SSA-SiamNet. (f) SFBS-FFGNET. (g) CSDBF. (h) CBANet. (i) Ours. (j) Ground truth.

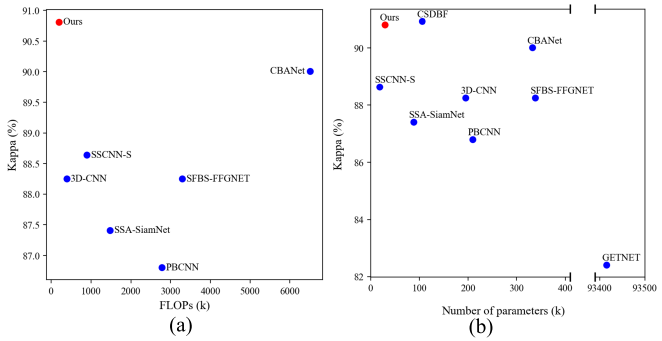

Fig. 10. Comparing performance versus model complexity in USA dataset. (a) Performance versus FLOPs. (b) Performance versus number of parameters.

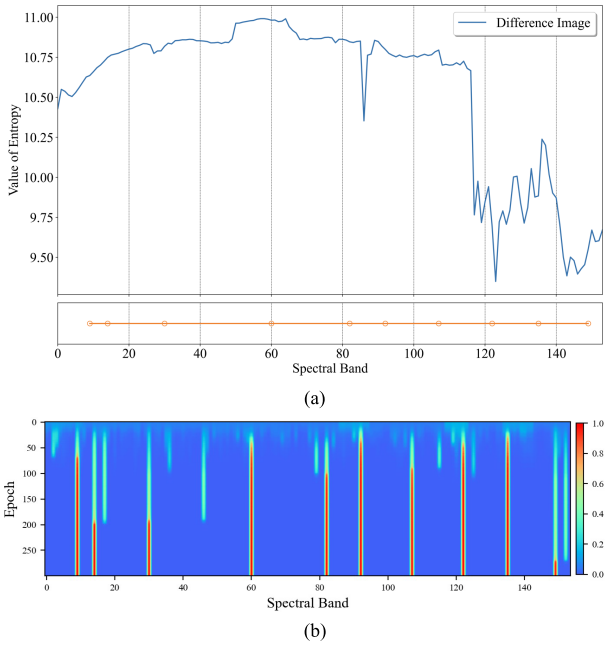

Fig. 11. Analysis of BS result in USA dataset. (a) Ten bands selected by our model (below) and the entropy value of each band (above). (b) Visualization of band importance weights across different epochs.

of model parameters, making it difficult for the model to converge with limited samples. Since our model achieves the best performance in the three datasets with an expansion rate of 3, we decided to set the number of channels in the three datasets to 36, 30, and 30, respectively. While these values may not be optimal, it provides generally good results across three datasets.

3) Patch Size: In patch-wise input networks, the patch size has a great influence on the CD performance. To this end, different patch sizes were used for experiments to analyze the influence of patch size on our model. As shown in Table V, when the patch size is set to 5×5, the detection performance is the best. The performance is the worst when the patch size is 9×9. In addition, we can observe that the performance rises and then falls as the patch size increases. This may be because the change information in 3×3 patches is accurate but not rich enough. And 7×7 or 9×9 patches contain too much peripheral information that interferes with the CD. Therefore, we finally set the patch size to 5×5.

G. Ablation Experiment

The proposed model ECDBS mainly consists of a BS module, BSA blocks, and a MF classifier. To demonstrate the effectiveness of the three modules, we fix all other parameters

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TABLE V
CD RESULTS OF DIFFERENT PATCH SIZES IN THREE DATASETS

Dataset	Patch Size	OA	Kappa	F1
River	3 × 3	97.29	82.17	82.74
	5 × 5	97.46	83.15	84.53
	7 × 7	97.17	80.17	81.70
	9 × 9	96.71	75.83	78.60
Farmland	3 × 3	98.52	95.85	96.82
	5 × 5	98.58	96.47	97.46
	7 × 7	98.13	94.81	96.03
	9 × 9	97.65	93.47	95.01
USA	3 × 3	96.48	88.82	91.00
	5 × 5	96.95	90.81	92.74
	7 × 7	96.30	88.13	90.43
	9 × 9	95.87	86.70	89.26

TABLE VI
ABLATION ANALYSIS IN THREE DATASETS

Dataset	Model	OA	Kappa	F1
River	ECDBS	97.46	83.15	84.53
	ECDBS-w/o BS	94.94	70.52	73.22
	ECDBS-w/o BSA	96.28	76.91	78.93
	ECDBS-w/o MF	97.34	82.41	83.85
Farmland	ECDBS	98.58	96.47	97.46
	ECDBS-w/o BS	97.43	93.64	95.44
	ECDBS-w/o BSA	98.33	95.84	97.00
	ECDBS-w/o MF	98.23	95.55	96.77
USA	ECDBS	96.95	90.81	92.74
	ECDBS-w/o BS	95.56	86.65	89.47
	ECDBS-w/o BSA	96.50	89.51	91.73
	ECDBS-w/o MF	96.61	89.68	92.00

to compare the CD results when varying the three modules. As shown in Table VI, a significant improvement can be observed in the application of BS module, demonstrating that a limited number of discriminative bands are sufficient to train a CD network. It can be observed that, on the basis of BS, BSA further improves model performance, with OA improving by 1.18%, 0.25%, and 0.45% in the three datasets, respectively. This is mainly due to the unique information of each band being highlighted by the tailored feature extraction strategy. Moreover, the MF classifier also shows a certain contribution to CD, with 0.12%, 0.35%, and 0.34% increase in OA. Overall, the experiments have shown that the BS module, the BSA block, and the MF classifier can effectively improve the CD performance.

H. Running Time Analysis

In addition to CD performance, running time is also an important index to measure the effectiveness of methods. For a fair comparison, we select some models that take patches as input and record the time required for testing and training one sample. As shown in Fig. 12, CBANet, 3D-CNN, and PBCNN exhibit short training and testing time, but this comes at the expense of CD performance. SSA-SiamNet achieves better performance and shows longer training time because the Siamese structure increases the computational cost in back propagation. However, our method not only achieves the best performance but also takes only 5.1 ms to train a sample and 1.12 ms to test a sample, which is superior to a lot of state-of-the-art methods. Overall, our model strikes a balance between running time and CD performance.

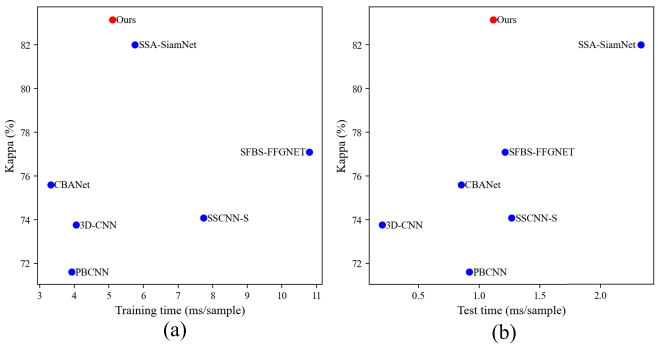

Fig. 12. Comparing performance versus running time in River dataset. (a) Performance versus training time. (b) Performance versus test time.

V. CONCLUSION

In this study, an ECDBS is proposed. The network incorporates a deep learning-based BS module as an internal unit, allowing us to jointly optimize BS and CD to exploit the synergy between them. ECDBS also includes cascaded BSA blocks to improve feature extraction. The BSA block adapts the feature strategy for each band based on its feature distribution, allowing features to be fully extracted from each band. We conducted comprehensive experiments on three benchmark datasets and compared the results with eight state-of-the-art HSI-CD algorithms, demonstrating the superiority of the proposed ECDBS. Finally, further research is needed to determine the optimal number of bands for CD. In this study, the number of selected bands is determined based on subjective experience. An adaptive method to determine this parameter will improve the generalization of this model.

REFERENCES

[1] G. M. Gandhi, S. Parthiban, N. Thummalu, and A. Christy, "NDVI: Vegetation change detection using remote sensing and GIS—A case study of Vellore district," *Proc. Comput. Sci.*, vol. 57, pp. 1199–1210, Jan. 2015.
[2] P. Washaya, T. Balz, and B. Mohamadi, "Coherence change-detection with Sentinel-1 for natural and anthropogenic disaster monitoring in urban areas," *Remote Sens.*, vol. 10, no. 7, p. 1026, Jun. 2018.
[3] J. Yan, L. Wang, W. Song, Y. Chen, X. Chen, and Z. Deng, "A time-series classification approach based on change detection for rapid land cover mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 249–262, Dec. 2019.
[4] P. Du, S. Liu, P. Gamba, K. Tan, and J. Xia, "Fusion of difference images for change detection over urban areas," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 4, pp. 1076–1086, Aug. 2012.
[5] H. Zhuang, K. Deng, H. Fan, and M. Yu, "Strategies combining spectral angle mapper and change vector analysis to unsupervised change detection in multispectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 681–685, May 2016.
[6] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.
[7] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.
[8] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.
[9] G. Hughes, "On the mean accuracy of statistical pattern recognizers," *IEEE Trans. Inf. Theory*, vol. IT-14, no. 1, pp. 55–63, Jan. 1968.

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[10] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716.
[11] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," *Proc. SPIE*, vol. 5133, pp. 326–336, Jan. 1980.
[12] F. Thonfeld, H. Feilhauer, M. Braun, and G. Menz, "Robust change vector analysis (RCVA) for multi-sensor very high resolution optical satellite data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 50, pp. 131–140, Aug. 2016.
[13] C. Kwan, "Methods and challenges using multispectral and hyperspectral images for practical change detection applications," *Information*, vol. 10, no. 11, p. 353, Nov. 2019.
[14] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.
[15] Z. Chen and B. Wang, "Spectrally–spatially regularized low-rank and sparse decomposition: A novel method for change detection in multitemporal hyperspectral images," *Remote Sens.*, vol. 9, no. 10, p. 1044, Oct. 2017.
[16] V. Ortiz-Rivera, M. Vélez-Reyes, and B. Roysam, "Change detection in hyperspectral imagery using temporal principal components," *Proc. SPIE*, vol. 6233, May 2006, Art. no. 623312.
[17] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, 1998.
[18] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for hyperspectral change detection," in *Proc. 6th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2014, pp. 1–4.
[19] D. R. Hardoon, S. Szedmak, and J. Shawe-Taylor, "Canonical correlation analysis: An overview with application to learning methods," *Neural Comput.*, vol. 16, no. 12, pp. 2639–2664, Dec. 2004.
[20] A. A. Nielsen, "The regularized iteratively reweighted mad method for change detection in multi-and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.
[21] L. Liu, D. Hong, L. Ni, and L. Gao, "Multilayer cascade screening strategy for semi-supervised change detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1926–1940, 2022.
[22] T. Abeywickrama, M. A. Cheema, and D. Taniar, "K-nearest neighbors on road networks: A journey in experimentation and in-memory implementation," *Proc. VLDB Endowment*, vol. 9, no. 6, pp. 492–503, Jan. 2016.
[23] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995.
[24] O. Ahlqvist, "Extending post-classification change detection using semantic similarity metrics to overcome class heterogeneity: A study of 1992 and 2001 US national land cover database changes," *Remote Sens. Environ.*, vol. 112, no. 3, pp. 1226–1241, Mar. 2008.
[25] T. Zhan et al., "SSCNN-S: A spectral–spatial convolution neural network with Siamese architecture for change detection," *Remote Sens.*, vol. 13, no. 5, p. 895, Feb. 2021.
[26] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510018.
[27] M. Gong et al., "A spectral and spatial attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521614.
[28] L. Wang, L. Wang, Q. Wang, and L. Bruzzone, "RSCNet: A residual self-calibrated network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529917.
[29] Y. Yang, J. Qu, S. Xiao, W. Dong, Y. Li, and Q. Du, "A deep multiscale pyramid network enhanced with spatial–spectral residual attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525513.
[30] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.
[31] Y. Li et al., "CBANet: An end-to-end cross-band 2-D attention network for hyperspectral change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513011.
[32] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.
[33] C. Shi, Z. Zhang, W. Zhang, C. Zhang, and Q. Xu, "Learning multiscale temporal–spatial–spectral features via a multipath convolutional LSTM neural network for change detection with hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529816.
[34] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.
[35] X. Wang, K. Zhao, X. Zhao, and S. Li, "CSDBF: Dual-branch framework based on temporal–spatial joint graph attention with complement strategy for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540118.
[36] P. Jian, Y. Ou, and K. Chen, "Uncertainty-aware graph self-supervised learning for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509019.
[37] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.
[38] X. Wang, K. Zhao, X. Zhao, and S. Li, "TriTF: A triplet transformer framework based on parents and brother attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507213.
[39] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016.
[40] Y. Wang, Q. Zhu, H. Ma, and H. Yu, "A hybrid gray wolf optimizer for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527713.
[41] W. Sun, J. Peng, G. Yang, and Q. Du, "Fast and latent low-rank subspace clustering for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3906–3915, Jun. 2020.
[42] J. W. P. Sun, H. Li, W. Li, X. Meng, C. Ge, and Q. Du, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Jun. 2021.
[43] Y. Zhou, Q. Yao, S. Huo, and X. Li, "Hyperspectral band selection with iterative graph autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511013.
[44] J. Feng, G. Bai, D. Li, X. Zhang, R. Shang, and L. Jiao, "MR-selection: A meta-reinforcement learning approach for zero-shot hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500320.
[45] H. Gao, Y. Zhang, Z. Chen, S. Xu, D. Hong, and B. Zhang, "A multidepth and multibranch network for hyperspectral target detection based on band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506818.
[46] C. Sui et al., "Unsupervised hyperspectral band selection with multigraph integrated embedding and robust self-contained regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504815.
[47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[48] A. Sharma, X. Liu, X. Yang, and D. Shi, "A patch-based convolutional neural network for remote sensing image classification," *Neural Netw.*, vol. 95, pp. 19–28, Nov. 2017.
[49] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.
[50] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018.

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